Module 04 • IoT & Embedded Edge

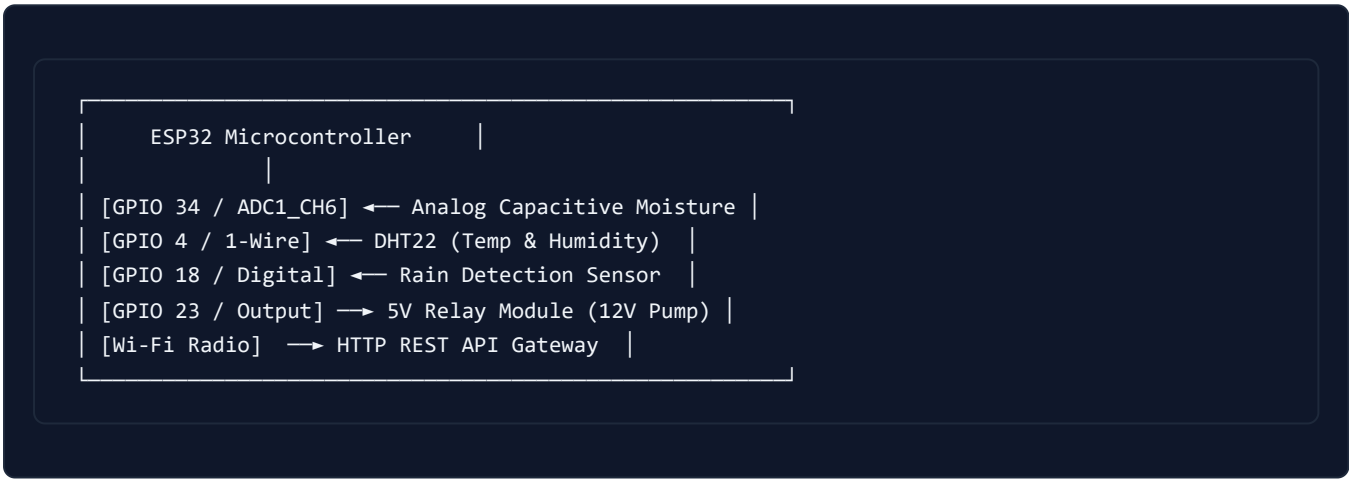
IoT Edge Architecture & ESP32 Integration

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

IoT Edge Architecture & ESP32 Integration

1. Microcontroller Hardware Topology

IRRIGATEWISE edge nodes utilize the **ESP32-WROOM-32D** dual-core microcontroller (Tensilica Xtensa 32-bit LX6, 240 MHz, 520 KB SRAM, integrated 2.4 GHz Wi-Fi).



Sensor Hardware Pinout:

Component	Sensor Pin	ESP32 Pin	Logic Level	Description
Capacitive Soil Sensor	VCC	3.3V	DC	Avoids electrolysis corrosion of resistive probes
	GND	GND	Ground	Common reference
	AOUT	GPIO 34	0–3.3V Analog	12-bit ADC reading (0 = Dry Air, 4095 = Submerged)
DHT22 Temperature/Hum	VCC	3.3V	DC	Digital 1-wire protocol
	DATA	GPIO 4	3.3V Logic	10k pull-up resistor
	GND	GND	Ground	
Rain Detection Sensor	DOUT	GPIO 18	3.3V Logic	Active Low (LOW = Precipitation detected)
12V Drip Pump Relay	IN	GPIO 23	3.3V Logic	Active High driving optocoupler isolation

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2. Telemetry Packet Schema

Edge devices serialize telemetry into standard JSON packets transmitted via HTTP `POST` `/api/v1/sensors/readings`:

```
"node_id": "ESP32-NODE-ERUMAIYUR-01",
"plot_id": "hydrogel_plot",
"soil_moisture": 48.65,
"temperature": 31.4,
"humidity": 68.2,
"rainfall": 0.0,
"hydrogel_presence": 1,
"hydrogel_dosage": 50.0,
"pump_status": "OFF",
"timestamp": "2026-09-20T11:30:00Z"
```

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3. Python ESP32 Edge Simulator CLI

To enable rapid prototyping, testing, and continuous integration without physical hardware attached, IRRIGATEWISE includes a feature-complete simulator: `backend/simulator/esp32_simulator.py`.

Simulator Features:

1. **Realistic Diurnal Simulation:** Simulates sinusoidal solar thermal cycles, humidity depression, and evapotranspiration.
2. **Dual-Plot Generation:** Alternates telemetry between `hydrogel_plot` (slow moisture desorption) and `control_plot` (rapid drying).
3. **Automated Dynamic Infiltration:** When pump status transitions to `ON`, subsequent cycles simulate moisture infiltration rising back to field capacity (62%).
4. **Fault Injection Mode:** Can inject synthetic sensor spikes, frozen ADC values, or pump-relay decoupling to test edge anomaly detection algorithms.

Running the Simulator:

```
# Transmit 48 continuous hourly cycles at 1-second acceleration
python backend/simulator/esp32_simulator.py --cycles 48 --interval 1.0

# Transmit continuous infinite stream to local server
python backend/simulator/esp32_simulator.py --url http://localhost:8000/api/v1/sensors/readings
```

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4. Anomaly Screening at Edge & Ingestion

Ingested packets are evaluated by `backend/ml/anomaly_detection.py` prior to database insertion: 1. **Physical Range Sanity:**

> **Equation:** `5.0% ≤ θ ≤ 95.0% and 0.0°C ≤ T ≤ 60.0°C`

2. **Instantaneous Spike Filter:**

> **Equation:** `|θ(t) - θ(t-1)| > 15.0% with Pump OFF ⇒ Outlier`

3. **ADC Freeze Detection:**

> **Equation:** `σ(θ_t-6:t) = 0.0 ⇒ Frozen ADC / Sensor Detached`

4. **Pump-Moisture Decoupling:**

> **Equation:** `Pump ON for > 20 min and Δθ ≤ 0 ⇒ Pump Air-Lock / Pipe Rupture`